Research and Sources Behind the Gennoor Tech Course Outlines

Compiled 28 October 2026. Every URL was opened during research and independently re-opened by a second reviewer to confirm the page loads and supports the stated fact. Citations are numbered in order of appearance; the reference list gives the full URL for each number.

How the outlines were built

1. **Client inputs.** *Some Use Cases. Yusuf.txt* (eleven automation use cases, each turned into a course named "<workflow> Automation with Claude") and *agentic_cybersecurity_course_prototypes.docx* (five product prototypes, each turned into a course). *AI-500 promo preview.mp4* set the visual language of the promotional videos and *Gennoor watermark process.zip* supplied the watermark procedure used on every PDF.
2. **Primary-source research.** For each course the vendor documentation, standards and official guides below were consulted to name the real APIs, commands, file locations and procedures that the modules teach.
3. **Verification.** Each citation was re-fetched and checked against the fact it supports; citations that did not load or did not contain the fact were dropped before this document was written.

Credential and Access Management Automation with Claude

- **Module 2: joiner access requests via ServiceNow Service Catalog REST API.** The Service Catalog API lives under the sn_sc namespace at /api/sn_sc/servicecatalog/; POST items/{sys_id}/order_now orders an item directly (sysparm_quantity is required, with optional variables and sysparm_requested_for) and returns the resulting sc_request number and sys_id, while POST cart/submit_order checks out the cart and behaves differently under one-step vs two-step checkout. **[1]**
- **Module 2: raising access requests through Jira Service Management.** POST /rest/servicedeskapi/request creates a customer request; the body must include serviceDeskId, requestTypeId and requestFieldValues covering every field the request type requires, needs the write:servicedesk-request scope, and returns 201 Created with the request object. **[2]**
- **Module 3: joiner provisioning in Microsoft Entra ID via Microsoft Graph.** POST /users creates an Entra ID user and requires accountEnabled, displayName, mailNickname, passwordProfile and userPrincipalName; the least-privileged permission for both delegated and application calls is User.Create, with User.ReadWrite.All and Directory.ReadWrite.All listed only as higher-privileged alternatives. **[3]**
- **Module 3: mover workflows - group membership changes via Microsoft Graph.** POST /groups/{group-id}/members/$ref adds one member with an @odata.id reference and PATCH /groups/{group-id} with members@odata.bind adds up to 20 at once; GroupMember.ReadWrite.All is the least-privileged permission, and newly created groups may briefly return 400 due to directory replication delay, so callers should retry after a short wait. **[4]**

- **Module 4: short-lived credentials and leaver revocation with HashiCorp Vault.** Vault attaches a lease with a TTL to every dynamic secret and service token; consumers must renew before expiry or the secret is automatically revoked, and revoking a lease (or a token, which revokes all leases it created, or an entire path prefix) invalidates the credential immediately, e.g. deleting the AWS access keys it issued. **[5]**
- **Module 5: least-privilege design for automated access grants.** NIST defines least privilege (per CNSSI 4009-2022, SP 800-53 Rev. 5 and SP 800-171r3) as restricting each user, process or entity to the minimum set of system resources and authorizations needed to perform its assigned function. **[6]**
- **Module 6: never placing secrets in LLM prompts (OWASP LLM Top 10).** OWASP lists security credentials among the sensitive data an LLM application can leak through its outputs, and recommends input validation and redaction before data reaches the model, least-privilege access, and tokenizing or masking sensitive content, warning that system-prompt restrictions alone can be bypassed via prompt injection. **[7]**
- **Module 6: secure MCP tool design for Claude-driven provisioning agents.** The MCP spec forbids token passthrough: an MCP server MUST NOT accept or forward tokens that were not explicitly issued to it, and it recommends a progressive least-privilege scope model that starts with minimal scopes and elevates via targeted WWW-Authenticate challenges rather than granting omnibus scopes up front. **[8]**
- **Module 6: hardening a Claude agent that touches identity systems.** Anthropic states it does not security-audit or manage any MCP server and advises writing your own or using trusted providers with per-server permissions; it also recommends not piping untrusted content directly to Claude, reviewing commands before approval, and notes API keys and tokens are kept in the OS keychain or a 0600-mode file rather than in prompts. **[9]**

Cisco ACI Upgrade Automation with Claude

- **Course foundation: primary upgrade/downgrade reference guide.** The guide is organized into an ACI Firmware Upgrade Overview, an Upgrade/Downgrade Architecture chapter, Pre-Upgrade/Downgrade Checklists, mixed-version operation guidance, and per-interface procedures (GUI by release, REST API, CLI). **[10]**
- **Module: end-to-end upgrade workflow and order (APIC cluster first, then switches).** Cisco's workflow upgrades the APICs first (after backup, image download, pre-upgrade validation and HUU/CIMC updates), then re-runs validations and upgrades the ACI-mode switches only once the APIC cluster is Fully Fit; multistep upgrades repeat the cycle through an intermediate release. **[11]**
- **Module: designing switch update groups.** Split leaf and spine switches into at least two update groups (e.g. ODD/EVEN) so that one member of every vPC pair, one APIC-connected leaf, and at least one route-reflector and IPN-connected spine per pod stay up; from APIC 4.2(5) the default concurrent capacity per group is unlimited, so Cisco recommends separate groups for scheduling. **[12]**
- **Module: pre-upgrade validation (built-in validator vs standalone script).** Three validation options exist: the built-in Pre-Upgrade Validator that runs automatically when an update group is configured, the now-obsolete App Center app (App Center deprecated in 6.1.2), and the standalone script run from the APIC CLI; Cisco advises running the script at least two weeks before the maintenance window, and from APIC 6.2(1) the script's external rules can be pulled into the Upgrade GUI via Intersight or manual download. **[13]**

- **Module: running and parsing the Pre-Upgrade Validation script with Claude.** The script is copied to /data/techsupport on an APIC, run with Python using admin credentials against a target version already present on the APIC, and reports each check as PASS, FAIL or MANUAL CHECK REQUIRED with a recommended action for each FAIL and a summary count by status. **[14]**
- **Module: validation reference and automating check triage.** The example output lists 47 checks (image/MD5, APIC cluster fitness, CIMC compatibility, SSD health, NTP, bootflash, overlapping VLAN pools, vPC pairing, L3Out MTU, ISIS metric for Multi-Pod) and the script saves results plus debug data in a .tgz bundle that can be attached to a TAC case. **[15]**
- **Module: driving the APIC upgrade via REST API.** A controller upgrade via REST is three POSTs: register the image source with firmwareRepoP/firmwareOSource at /api/node/mo/uni/fabric.xml, set the target version with firmwareCtrlrFwP at /api/node/mo/uni/controller.xml, then trigger it with maintCtrlrMaintP adminState="up" adminSt="triggered". **[16]**
- **Module: APIC REST API authentication and session handling for automation.** Clients POST to aaaLogin (prefixing the username with apic:domain\ when multiple login domains exist), receive a token in the APIC-cookie Set-Cookie header and aaaLogin token attribute with a default refreshTimeoutSeconds of 600, and must call aaaRefresh before expiry, storing the new token returned on every refresh. **[17]**
- **Module: reading release notes and compatibility matrices before a target version is chosen.** Release 6.1(4h) became available on August 13, 2025; the notes instruct installing only the CIMC versions listed in the compatibility table and reviewing software advisory CSCwo74485 for UCS C225 M6 (APIC-L4/M4) servers before upgrading, alongside new features such as virtual APIC on Nutanix AHV and TACACS+ over TLS. **[18]**

Bill Creation Automation with Claude

- **Module 1: GST tax invoice vs bill of supply and Rule 46 mandatory fields.** GST law prescribes no fixed invoice layout, but Rule 46 of the CGST Rules 2017 makes a set of fields mandatory; a registered person making taxable supplies issues a tax invoice, while composition dealers or exempt-only suppliers issue a bill of supply, and no invoice is needed for supplies under Rs. 200 subject to conditions. **[19]**
- **Module 2: pulling vendor bills from an Outlook/M365 mailbox via Graph.** GET /me/messages (or /users/{id}/messages) returns 10 messages per page by default, $top can raise it to 1000, Mail.ReadBasic is the least-privileged permission, and paging must follow the @odata.nextLink URL rather than hand-building $skip values. **[20]**
- **Module 2: reading a single bill email body as plain text or raw MIME.** GET /me/messages/{id} returns the message object; appending /$value returns the raw MIME content instead, and sending Prefer: outlook.body-content-type="text" returns the body as plain text rather than the default HTML. **[21]**
- **Module 2: finding invoice emails in Gmail with search queries.** messages.list returns only message IDs and thread IDs, so each message must then be fetched with messages.get; the q parameter accepts Gmail search syntax such as from:someuser@example.com is:unread to narrow results. **[22]**
- **Module 2: downloading PDF bill attachments from Gmail.** Attachments are fetched with a GET on users/{userId}/messages/{messageId}/attachments/{id} with an empty request body, returning the attachment data for downstream extraction. **[23]**

- **Module 3: sending supplier invoice PDFs to Claude for extraction.** Claude accepts standard unencrypted PDFs up to 32 MB per request and up to 600 pages (100 pages when the context window is under 1M tokens), and can read text, tables and charts to convert them into structured data. **[24]**
- **Module 3: schema-constrained JSON for invoice line items, GSTIN and tax totals.** Setting output_config.format to type json_schema with a schema constrains Claude's response to that schema so it can be parsed without retry loops; output may still diverge only on refusals (stop_reason refusal) or max_tokens truncation, and the older output_format parameter is deprecated. **[25]**
- **Module 4: rendering the HTML bill template to a PDF file.** page.pdf() renders the page using the print CSS media type, so colours are adjusted for printing by default; call page.emulateMediaType('screen') first to render the on-screen styling instead. **[26]**

Certificate Renewal Automation with Claude

- **Module: F5 BIG-IP certificate install via iControl REST.** The /mgmt/tm/sys/file/ssl-cert collection supports GET; individual certificate resources support GET, POST, PUT, PATCH and DELETE, are keyed by name, partition and subPath, and take an optional sourcePath property that names the URI the certificate file is copied from. **[27]**
- **Module: NetScaler certificate install via NITRO API.** Creating a certificate-key pair is a POST to /nitro/v1/config/sslcertkey with a JSON body whose mandatory fields are certkey and cert (paths point to files on the appliance disk, typically /nsconfig/ssl); a successful add returns 201, and UPDATE is a PUT to the same URL with certkey mandatory and optional fields such as expirymonitor and notificationperiod. **[28]**
- **Module: NetScaler certificate binding to SSL vservers.** After installing the cert and key under /nsconfig/ssl you run add ssl certKey <name> -cert <file> -key <file> (optionally -expiryMonitor ENABLED -notificationPeriod <days>), link any intermediate certificates first, then bind with bind ssl vserver <vs> -certkeyName <name> and verify with show ssl vserver; every SSL virtual server needs its own valid binding. **[29]**
- **Module: Radware Alteon certificate management.** On Alteon the installation flow is to build one PEM file with the server certificate followed by the intermediate chain, enter the cfg then cert menu on the switch CLI, paste the PEM, terminate the input with three periods, and then run apply to activate the change (no official Radware public page could be fetched, so this CA guide is the best available source). **[30]**
- **Module: ACME and internal CA renewal workflow.** ACME replaces manual Web PKI steps with machine-readable exchanges: a client creates an account, submits an order for identifiers, satisfies one of the challenges offered in each authorization to prove control, finalizes the order with a CSR and then downloads the issued certificate; the protocol also covers revocation. **[31]**
- **Module: OpenSSL verification of installed chains.** openssl verify validates certificate chains and tries to report every problem it finds rather than stopping at the first; -CAfile supplies trusted anchors while -untrusted (repeatable) supplies intermediates used only for chain building, not as trust anchors. **[32]**
- **Module: building an MCP server for load balancer APIs.** MCP is an open-source standard, likened to a USB-C port for AI applications, that lets AI apps connect to data sources, tools and workflows in a uniform way; it is supported by Claude, ChatGPT, VS Code, Cursor and others so a server built once works across clients. **[33]**

- **Module: orchestrating renewals with Claude Code.** Claude Code is an agentic coding tool available in terminal, IDE, desktop app and browser; it can be scripted non-interactively with claude -p for piping and CI use, connects to external systems through MCP, supports hooks that run shell commands before or after its actions, and can run cloud or desktop scheduled tasks for recurring work. **[34]**
- **Module: building a renewal agent with the Claude Agent SDK.** The Agent SDK exposes the same tools, agent loop and context management as Claude Code as a Python or TypeScript library, with built-in tools, hooks, subagents, MCP connections, permission controls, resumable sessions, skills and plugins; third-party products must use API-key authentication rather than claude.ai logins. **[35]**

Configuration Comparison Automation with Claude

- **Module 1: parsing BIG-IP virtual server objects from bigip.conf / tmsh list.** A BIG-IP virtual server is defined by properties such as destination (address:port), pool, profiles, persist, rules, policies, vlans and source-address-translation; `list virtual` prints the configuration while `show virtual` adds runtime statistics, and the older snat/snatpool options are deprecated in favor of source-address-translation. **[36]**
- **Module 2: extracting and diffing NetScaler ns.conf (running vs saved config).** NetScaler stores its saved configuration in /nsconfig/ns.conf; `save ns config` persists changes, `show ns runningConfig` and `show ns ns.conf` display running and saved config, and the built-in `diff ns config <file1> <file2>` command (or GUI Configuration difference) compares two configuration files. **[37]**
- **Module 5: NetScaler config converter and instance-to-instance migration.** The Configuration Builder accepts a live instance, pasted ns.conf CLI commands, or an uploaded config file, generates a StyleBook plus a reusable config pack for migrating to other instances, and does not detect named expressions or responder HTML pages, which must be pre-created on the target. **[38]**
- **Module 3: mapping ADC objects to AWS ELB listeners, rules and target groups.** An ALB is composed of listeners (protocol/port), listener rules (priority, conditions, actions, with a mandatory default rule) and target groups that route to registered targets with per-target-group health checks; a target may belong to multiple target groups and the default routing algorithm is round robin. **[39]**
- **Module 3: mapping ADC objects to Azure Application Gateway components.** Azure Application Gateway routes traffic through frontend IPs, listeners (basic or multi-site, up to 5 host names each), request routing rules (basic or path-based, one listener per rule), HTTP/backend settings, backend pools and health probes; HTTP/2 is client-side only and backend communication is always HTTP/1.1. **[40]**
- **Module 4: F5 migration tooling, converting imperative config to AS3/DO declarations.** BIG-IP ACC converts existing BIG-IP configuration files into declarative AS3 (application/service) or Declarative Onboarding (system, network, clustering) JSON declarations, and is a community-supported tool whose source and issue tracker live on GitHub. **[41]**
- **Module 4: F5 Journeys UCS-based version and platform migration.** F5 Journeys (successor to the BIG-IP Migration Assistant, code at github.com/f5devcentral/f5-journeys) accepts or generates a UCS archive, migrates full configurations from 11.5.0+ to newer releases including VELOS and rSeries, flags feature-parity gaps with built-in fixes, and can extract individual applications into AS3. **[42]**

- **Module 6: building a Claude-driven comparison agent with tool calls.** To let Claude invoke a custom function (for example a config parser or diff tool), you pass a tool with a JSON input_schema; Claude responds with stop_reason tool_use and a tool_use block, your code executes it and returns a tool_result block, and `strict: true` guarantees calls conform to the schema. **[43]**
- **Module 7: semantic diff approaches and drift detection patterns.** terraform plan refreshes state, compares desired configuration against prior state and reports a semantic diff using + create, - destroy, ~ update in-place and -/+ replace markers, with -detailed-exitcode returning 0 (no changes), 1 (error) or 2 (changes present) for automation pipelines. **[44]**

Traffic and Flow Diagram Automation with Claude

- **Module 1: traceroute fundamentals and the IP Traceroute option.** Classic TTL-based traceroute needs roughly 2n packets for n hops; RFC 1393 proposed an IP Traceroute option plus an ICMP Traceroute message (type 30) so each router reports back directly, cutting this to about n+1 packets and avoiding mid-trace path shifts. It is Experimental and was later obsoleted by RFC 6814. **[45]**
- **Module 1: ping fundamentals (ICMP Echo/Echo Reply) and Time Exceeded.** ICMP Echo is type 8 and Echo Reply is type 0, with identifier and sequence fields used to match replies to requests and the received data echoed back unchanged; Time Exceeded is type 11, with code 0 meaning TTL exceeded in transit, which is the message traceroute relies on. **[46]**
- **Module 1: how traceroute probes work on Cisco IOS, Linux and Windows.** Cisco IOS and Linux traceroute send UDP probes starting at destination port 33434 with TTL starting at 1 (default max 30 hops); each hop returns ICMP Time Exceeded (type 11, code 0) and the destination returns ICMP Port Unreachable (type 3, code 3). Windows tracert uses ICMP Echo instead, and Cisco routers rate-limit ICMP unreachables (default one per 500 ms), which can show up as missing hops. **[47]**
- **Module 2: continuous path measurement with mtr.** mtr combines traceroute and ping in one tool: it sends probes with successively larger TTL values to discover the path and then prints running per-hop statistics. The README warns that routers may not respond to or may rate-limit ICMP, so apparent loss at an intermediate hop does not necessarily mean forwarded traffic is being dropped there. **[48]**
- **Module 3: Mermaid flowchart syntax for network path diagrams.** Flowcharts start with `flowchart` (or `graph`) plus a direction keyword (TB/TD, BT, RL, LR); edges include `⟶`, `---`, dotted `-.-`, thick `===`, and `--o`/`--x`; `subgraph id [Title] ... end` groups nodes and can carry its own direction; from v11.3.0 shapes can be set with `A@{ shape: ... }`. A lowercase `end` inside a node label breaks the diagram. **[49]**
- **Module 3: rendering Claude-generated Mermaid in GitHub Markdown.** GitHub renders a Mermaid diagram from a fenced code block labeled `mermaid` in issues, discussions, pull requests, wikis and Markdown files; placing `info` in a mermaid block reveals which Mermaid version GitHub is running. **[50]**
- **Module 4: draw.io CSV import for automated topology diagrams.** Via Arrange > Insert > Advanced > CSV, configuration lines starting with `#` precede the CSV data; `# label:` builds shape text from `%column%` placeholders, `# connect:` takes a JSON object with `from`/`to` columns (plus label, style, invert), `# identity:` lets a re-import update existing shapes instead of duplicating them, and `# layout: auto` with spacing keys applies an automatic layout. **[51]**

- **Module 4: Graphviz DOT for programmatic path graphs.** A DOT file declares `graph` (undirected, edges `--`) or `digraph` (directed, edges `->`), optionally prefixed with `strict` to forbid duplicate edges; attributes are `key=value` pairs in square brackets, `graph`/`node`/`edge` statements set defaults only for objects defined afterwards, and subgraph names beginning with `cluster` are drawn as grouped bounding boxes by some layout engines. **[52]**
- **Module 5: prompting Claude to emit clean Mermaid/DOT output.** To control output format, tell Claude what to do rather than what not to do, wrap content types in XML tags (e.g. instructions, context, input) to reduce misinterpretation, and use XML format indicators or structured outputs/tool calling for a fixed structure; the page also notes that few-shot examples are one of the most reliable ways to steer output format and structure. **[53]**

Device Health Monitoring Automation with Claude

- **Module: NetScaler virtual server health via NITRO stat API.** GET /nitro/v1/stat/lbvserver (or /lbvserver/<name>) returns per-vserver state values such as UP(7), DOWN(1), OFS(4), TROFS(5), plus vslbhealth (percentage of bound services that are UP), actsvcs and totalrequests. **[54]**
- **Module: NetScaler HA pair and config-sync status checks.** GET /nitro/v1/stat/hanode exposes hacurmasterstate (PRIMARY/SECONDARY/CLAIMING/FORCE CHANGE), hacurstate (UP, DISABLED, PARTIALFAIL, COMPLETEFAIL, etc.) and haerrsyncfailure, a counter of failed primary/secondary config syncs since the last transition, commonly caused by an RPC password mismatch. **[55]**
- **Module: F5 BIG-IP polling via iControl REST.** iControl REST is the REST successor to the SOAP iControl framework; anything doable from the BIG-IP command line can be done programmatically, with LTM objects such as pools reached under /mgmt/tm/ltm/ (example: POST to /mgmt/tm/ltm/pool with basic-auth credentials). **[56]**
- **Module: interpreting F5 pool member state/session fields.** Pool member objects expose a state field (user-down forces the member offline overriding monitors; user-up returns it to its monitor-derived state), a session field (default user-enabled, controlling new sessions) and a monitor field listing assigned health monitors and their status. **[57]**
- **Module: ServiceNow incident creation via Table API.** POST /now/table/{tableName} inserts exactly one record (bulk insert not supported), accepts JSON or XML bodies via Content-Type, returns 201 with a Location header for the new record, and the calling user must hold roles sufficient to access the target table; an example posts short_description, assignment_group, urgency and impact to incident. **[58]**
- **Module: hands-on lab creating an incident from a health-check failure.** The walkthrough POSTs to https://instance.service-now.com/api/now/v1/table/incident with a body like {"short_description":"...","comments":"..."}; it requires the admin, web_service_admin or rest_api_explorer role and the response includes a Location header pointing at the new incident. **[59]**
- **Module: scheduling health checks with GitHub Actions cron.** The schedule event uses five-field POSIX cron in UTC (a time zone can be set), minimum interval is every 5 minutes, runs only from the default branch, can be delayed at the top of the hour under load, and scheduled workflows in public repos are auto-disabled after 60 days of inactivity. **[60]**
- **Module: orchestrating device checks and ticketing with a Claude agent.** The Claude Agent SDK, available in Python and TypeScript, runs the same agent loop as Claude Code with built-in

tools (file read/write, run commands, web search), hooks, subagents, permissions, sessions and MCP support for connecting external tools and data sources. **[61]**

- **Module: Radware Alteon monitoring and REST API reference pointer.** Radware's official Alteon documentation set includes a dedicated 'Alteon REST API User Guide' alongside the Command Reference, CLI Application Guide and Troubleshooting Guide, which are the primary references for programmatic status and uptime checks on Alteon ADCs. **[62]**

HR Hiring Flow Automation with Claude

- **Module 2: exporting candidates and applications from an ATS via API.** Harvest was built so customers can export their Greenhouse data; it uses Basic Auth over HTTPS with the API token as the username and blank password, exposes list/retrieve/add/edit/anonymize endpoints for candidates and advance/move/reject/hire endpoints for applications, and Harvest v1/v2 is deprecated for removal on 31 August 2026 in favour of v3. **[63]**
- **Module 2: pulling candidate pipeline data from Lever.** The Lever Data API is JSON over HTTPS at https://api.lever.co/v1, authenticates with Basic Auth API keys or OAuth 2.0 (access tokens expire after one hour), is rate limited to 10 requests/second per key with bursts to 20, and the Candidates endpoints are deprecated since 2020 in favour of Opportunities. **[64]**
- **Module 2: LinkedIn Talent API integration options for an ATS.** LinkedIn Talent Solutions offers partner-facing APIs grouped as Apply Connect, Apply with LinkedIn, Job Posting API, CRM Connect and Recruiter System Connect; Recruiter System Connect synchronises candidate information between an ATS and LinkedIn Recruiter, while the Job Posting API ingests jobs already created in an ATS or job board. **[65]**
- **Module 2: Naukri job-posting integration and its access constraints.** Naukri has no self-serve developer portal: an employer must request its team's API key from its Naukri point of contact, Naukri requires a paid subscription to post jobs (which may cap active postings), and postings fail unless location, experience range and job role are present and correctly formatted. **[66]**
- **Module 5: lawful basis, notice and consent for candidate data under DPDP Act 2023.** Section 5 requires every consent request to be accompanied or preceded by a notice stating the personal data and purpose, how to exercise rights and how to complain to the Board; Section 6 requires consent to be free, specific, informed, unconditional and unambiguous with a clear affirmative action; and Section 7(i) lists processing for the purposes of employment as a 'certain legitimate use' that does not require consent. **[67]**
- **Module 5: compliance timeline, consent notices, breach notification and penalties.** The DPDP Rules, 2025 were notified on 14 November 2025 with an eighteen-month phased compliance period; every Data Fiduciary must issue a separate, clear, purpose-specific consent notice, Consent Managers must be India-based companies, breaches must be reported to affected individuals and the Data Protection Board, and penalties run up to Rs 250 crore for failing to maintain reasonable security safeguards. **[68]**
- **Module 3: scoring candidate English proficiency on the CEFR scale.** CEFR has six levels grouped into three bands: Basic user (A1 Breakthrough, A2 Waystage), Independent user (B1 Threshold, B2 Vantage) and Proficient user (C1 Effective operational proficiency, C2 Mastery); B2 is where a candidate can follow complex technical text in their field and converse fluently with native speakers, and levels can be subdivided locally (for example A1.1/A1.2). **[69]**
- **Module 4: schema-constrained screening output from Claude.** Passing output_config.format with type json_schema makes Claude return JSON that matches the schema (no beta header needed; the older output_format parameter is deprecated); objects must set

additionalProperties:false, recursive schemas and numeric min/max constraints are unsupported, and strict:true on a tool can be combined in the same request, with schemas cached for 24 hours so sensitive data must not be put in schema definitions. **[70]**

- **Module 4: wiring ATS and portal APIs to Claude as tools.** A client tool is defined with a name, description and JSON input_schema; Claude answers with stop_reason tool_use and a tool_use block, the application runs the call and returns a tool_result block in the next user turn, and tool_choice (auto by default, with disable_parallel_tool_use available) controls whether and how Claude must call a tool, while strict:true guarantees the inputs match the schema. **[71]**

Terraform and GitHub Automation with Claude

- **Module 1: Terraform fundamentals - state and common mistakes.** Terraform maps real infrastructure to configuration through state stored per workspace; state is a JSON file that should not be hand-edited or committed to version control, and HashiCorp recommends a remote backend or HCP Terraform instead of the default local file. **[72]**
- **Module 1: remote backends and shared state.** A configuration may contain only one backend block; backend settings are written in plain text to .terraform/terraform.tfstate and plan files, so credentials should be supplied via environment variables rather than in the configuration. **[73]**
- **Module 1: Terraform modules.** A module is a collection of resources Terraform manages together; modules can be sourced from the local filesystem, a Terraform registry, or VCS repositories, and Terraform downloads registry modules automatically when source and version are set in a module block. **[74]**
- **Module 1: terraform fmt/validate in CI.** terraform fmt rewrites files to a single canonical style with no customization options; the -check flag returns a non-zero exit status when files are not formatted, and -recursive is needed to process subdirectories, which makes it suitable as a CI gate. **[75]**
- **Module 2: branches, pull requests and branch protection rules.** Branch protection rules can require passing status checks or a linear history before pushes, block force pushes and deletions by default, and do not apply to admins unless 'Do not allow bypassing the above settings' is enabled; only one branch protection rule applies to a branch at a time, a limitation rulesets do not share. **[76]**
- **Module 2: CODEOWNERS for Terraform review ownership.** A CODEOWNERS file placed in .github/, the repo root, or docs/ automatically requests review from the listed owners when a PR touches their files; the file on the PR's base branch is used, owners must have write access, invalid lines are skipped, and files over 3 MB are not loaded. **[77]**
- **Module 3: GitHub Actions workflows for Terraform plan/apply.** The reference workflow runs terraform plan on every commit to a pull request branch so changes can be reviewed before merge, then runs apply when the PR is merged to main; it authenticates to HCP Terraform with a team token stored as the TF_API_TOKEN GitHub secret. **[78]**
- **Module 4: F5 BIG-IP Terraform provider and AS3 declarations.** The bigip_as3 resource takes a required as3_json argument pointing to an AS3 declaration file and an optional tenant_filter that restricts changes to a single tenant; it requires BIG-IP 12.1.x or later, AS3 extension 3.10 or later, and an Administrator-role account. **[79]**
- **Module 5: Claude Code GitHub Actions for Terraform PR automation.** The anthropics/claude-code-action runs in interactive mode (responding to @claude mentions in issues and PRs) when no prompt input is given, or in automation mode on any GitHub event when a prompt is supplied;

it authenticates via an ANTHROPIC_API_KEY or CLAUDE_CODE_OAUTH_TOKEN secret, requires the triggering user to have write access, and rejects bot actors unless listed in allowed_bots. **[80]**

Troubleshooting Automation with Claude

- **Module: creating and updating incidents via the ServiceNow Table API.** The Table API exposes CRUD endpoints under /now/table/{tableName}: GET lists or reads records, POST inserts a single record (no bulk insert), PATCH/PUT update by sys_id, and DELETE returns HTTP 204 on success; a versioned path /api/now/{api_version}/table/... is optional and defaults to the latest version. **[81]**
- **Module: incident email notifications and acknowledgement replies.** Every outbound notification carries a watermark label that lets ServiceNow match an inbound reply back to the originating record, and inbound email actions define how those replies (for example an acknowledgement) are processed; notifications can also be batched into digests over a time interval. **[82]**
- **Module: automating incident workflows with Flow Designer.** A flow starts from a record-, schedule- or application-based trigger; matching trigger events are placed in an event queue and processed in order by a scheduler, so a flow may start with a delay, after which the engine builds a cached, just-in-time compiled process plan of published actions and subflows. **[83]**
- **Module: F5 BIG-IP troubleshooting with tmsh.** The TMSH reference is the collected BIG-IP TMOS Shell man pages organized into General, Commands and Modules sections, and which man pages apply depends on which modules are provisioned on the BIG-IP system. **[84]**
- **Module: Citrix NetScaler troubleshooting with show commands.** The NetScaler ADC CLI command reference is indexed by feature area (aaa, lb, ssl, system, vpn and so on), with a commands page per area and sub-pages for individual configuration objects such as lb-vserver and ssl-certKey. **[85]**
- **Module: Cisco IOS/NX-OS connectivity troubleshooting.** The extended ping command on Cisco IOS is available only from privileged EXEC mode, whereas the basic ping works in both user EXEC and privileged EXEC modes. **[86]**
- **Module: knowledge base templates for troubleshooting runbooks.** ServiceNow ships Standard, FAQ, How To, What Is and KCS Article templates (the KCS template has Issue, Environment, Cause and Resolution fields); all but Standard are inactive by default, a template must be active and linked to a knowledge base to appear to authors, and the feature requires the Knowledge Management Advanced plugin. **[87]**
- **Module: wiring Claude Code to ServiceNow and network devices via MCP servers.** Servers are registered with claude mcp add using HTTP (recommended for remote), stdio (local process) or deprecated SSE transports, and stored at local, project (.mcp.json, shareable via version control) or user scope; when a server is defined in several places, local takes precedence over project, then user, then plugin and claude.ai connector definitions. **[88]**
- **Module: MCP fundamentals for building troubleshooting agents.** MCP is an open-source standard for connecting AI applications to external data sources, tools and workflows, described as a USB-C-style standardized port so a server can be built once and used across clients such as Claude, VS Code and Cursor. **[89]**

Device Upgrade Automation with Claude

- **Module: F5 BIG-IP software image install (tmsh install sys software image).** The ISO must be placed in /shared/images and installed to an inactive volume with `tmsh install sys software image <iso> volume HD1.x`; progress is watched with `tmsh show sys software` and logged to /var/log/liveinstall.log. A UCS backup must be saved first, mirroring turned off, and on redundant pairs the standby unit is upgraded and verified before the active unit. **[90]**
- **Module: F5 BIG-IP HA / device group upgrade sequence and pre-checks.** Each device in a device group is forced offline (moving its traffic groups to a peer) before it is upgraded, one device at a time; mirroring does not function until every device has rebooted onto the new version, so F5 tells you to do this inside a maintenance window. Preparation includes config sync, RAID health check, QKView to iHealth, UCS backup, and image checksum verification. **[91]**
- **Module: Citrix NetScaler HA pair upgrade order.** Upgrade the secondary node first, force a failover so it becomes primary, then upgrade the former primary. While the two nodes run builds with different internal HA versions, config sync, command propagation, connection mirroring, and persistence-session sync are disabled; `show ha node` should be run beforehand to confirm the node state. **[92]**
- **Module: NetScaler disk space and pre-upgrade validation checks.** NetScaler 14.1 requires at least 7 GB free in /var (1 GB bundle, 2 GB extraction, 4 GB upgrade process) and at least 250 MB free in /flash. The page also says to upgrade one major release at a time, allocate roughly twice the normal time, back up config/certs/licenses, remove /var/customizations, run the preconfiguration check tool, run fsck on hardware, and test in non-production first. **[93]**
- **Module: Radware Alteon software upgrade path, pre-upgrade steps and rollback.** Before upgrading, back up the configuration and run the Upgrade Advisor Tool (from the Radware Customer Portal, with its database updated) against the current config and target version. Versions 31.x-33.x upgrade directly, older ones need intermediate hops; configuration rollback is not supported, so downgrading means booting the old image with factory config and re-uploading the saved configuration. **[94]**
- **Module: Radware device software update workflow (File > Software Update).** Radware's device update flow requires an upgrade password obtained from Radware support that is tied to the device's base MAC address and the software file size; after backing up the configuration, the operator enters the version and file, chooses whether to enable the new version immediately, applies the update, and the device reboots and must be verified on the console. **[95]**
- **Module: scheduling and framing upgrades as routine preventive maintenance.** Published April 2022, SP 800-40r4 defines enterprise patch management as identifying, prioritizing, acquiring, installing, and verifying patches, updates, and upgrades, and frames it as preventive maintenance that organizations should operationalize with an enterprise-wide strategy rather than treat as ad hoc work. **[96]**

Agentic Virtual Patching Platform

- **Module 1: virtual patching lifecycle and best practices.** OWASP defines virtual patching as an enforcement layer that blocks exploitation of a known flaw without changing the vulnerable code, and lays out a six-phase process (Preparation, Identification, Analysis, Patch Creation, Implementation/Testing, Recovery/Follow-Up) that recommends deploying new rules in log-only mode first and retesting with alternate encodings before trusting them. **[97]**

- **Module 2: OWASP CRS anomaly scoring and paranoia levels.** CRS separates detection from blocking: each matching rule adds points by severity (CRITICAL 5, ERROR 4, WARNING 3, NOTICE 2) and the request is only denied when the summed inbound score reaches the threshold, which defaults to 5; per-paranoia-level scores are added together before the blocking check. **[98]**
- **Module 2: writing SecRule virtual patches and staged rollout with DetectionOnly.** SecRuleEngine accepts On, Off (the default) and DetectionOnly; in DetectionOnly rules are evaluated and logged but disruptive actions such as deny, drop and redirect are never executed, which is the mechanism for a safe pre-block canary stage. SecRule syntax is SecRule VARIABLES OPERATOR [ACTIONS], with rule actions merged over the default action list. **[99]**
- **Module 3: deploying virtual patches on NGINX WAF.** NGINX App Protect WAF is now branded F5 WAF for NGINX; it can be installed on VMs or bare metal, Kubernetes, Docker and air-gapped environments, and its policies combine attack signatures, bot signatures, threat campaigns, response and server-technology signatures plus user-defined signatures, with separate guidance for signature updates. **[100]**
- **Module 4: CVE triage and severity thresholds for auto-patching.** CVSS v4.0 has Base, Threat, Environmental and Supplemental metric groups (Supplemental metrics never change the score), and maps numeric scores to severity bands: Low 0.1-3.9, Medium 4.0-6.9, High 7.0-8.9, Critical 9.0-10.0. **[101]**
- **Module 4: prioritizing CVEs with the KEV catalog.** CISA describes the KEV catalog as the authoritative list of vulnerabilities exploited in the wild (about 1,760 entries at time of access), publishes it as CSV and JSON for automation, and tells organizations to use it as an input to their vulnerability-management prioritization alongside vendor mitigations and asset internet-exposure assessment. **[102]**
- **Module 5: SLA-driven approval and remediation timelines.** Under BOD 26-04 remediation urgency is driven by four factors (public exposure, KEV listing, automatable exploitation, technical impact); the clock starts when CISA adds a CVE to KEV or the agency detects it on an asset, and cases marked for forensic triage must be mitigated within three days with the asset checked for compromise. **[103]**
- **Module 6: agent safety risks (prompt injection, excessive agency).** The 2025 list places Prompt Injection at LLM01 and Excessive Agency at LLM06, alongside Improper Output Handling (LLM05) and Unbounded Consumption (LLM10), which together frame the controls an autonomous patching agent needs around untrusted CVE text and tool permissions. **[104]**
- **Module 7: building the patching agent with Claude tool use.** Client tools return a stop_reason of tool_use with tool_use blocks that your application executes and answers with tool_result blocks, while server tools run on Anthropic infrastructure; tool_choice can force or disable parallel calls and strict: true guarantees schema-conformant tool inputs. **[105]**
- **Module 7: human-in-the-loop approval gates for agent tools.** The MCP spec's Security and Trust & Safety section states that tools represent arbitrary code execution, that hosts must obtain explicit user consent before invoking any tool, and that tool descriptions and annotations should be treated as untrusted unless they come from a trusted server. **[106]**

Agentic AI Vulnerability Management

- **Module 2: scoring with CVSS v4.0.** CVSS v4.0 defines four metric groups (Base, Threat, Environmental, Supplemental) and requires scores to be labeled by which groups were used: CVSS-B, CVSS-BT, CVSS-BE or CVSS-BTE; the Base score ranges 0.0 to 10.0. **[107]**

- **Module 2: CVSS v3.1 severity bands.** CVSS v3.1 uses Base, Temporal and Environmental metric groups, and maps scores to qualitative ratings: None 0.0, Low 0.1-3.9, Medium 4.0-6.9, High 7.0-8.9, Critical 9.0-10.0. **[108]**
- **Module 3: exploit likelihood with EPSS.** EPSS is a machine-learning model that outputs, daily for every published CVE, a 0-1 probability (plus percentile) that the CVE will be exploited in the wild within the next 30 days; data is free via CSV download and an API. **[109]**
- **Module 3: KEV-driven prioritization.** CISA positions the KEV catalog as the authoritative source of vulnerabilities exploited in the wild and publishes it as CSV and JSON with a JSON Schema; each entry carries a date added, a due date and a known-ransomware-use flag. **[110]**
- **Module 3: KEV inclusion criteria and remediation SLAs.** A vulnerability enters the KEV catalog only if it has a CVE ID, clear remediation guidance and reliable evidence of in-the-wild exploitation; the directive set two-week remediation for newer CVEs and six months for pre-2021 CVEs, and has since been superseded by BOD 26-04. **[111]**
- **Module 7: patch management process per NIST SP 800-40.** SP 800-40 Rev. 4 (final, April 2022, superseding Rev. 3 from 2013) frames patching as routine preventive maintenance and recommends an enterprise-wide patching strategy to operationalize the process and reduce risk. **[112]**
- **Module 4: ingesting Trivy JSON and SARIF exports.** Trivy's --format flag supports table (default), JSON, SARIF (2.1.0), template, SBOM and GitHub dependency snapshot; in JSON output VulnerabilityID, PkgName, InstalledVersion and Severity are always populated while other fields may be empty. **[113]**
- **Module 4: SARIF format constraints for scanner normalization.** GitHub code scanning accepts only SARIF 2.1.0, limits each gzip-compressed upload to 10 MB and each run to 25,000 results (top 5,000 retained), and rejects empty strings for required properties. **[114]**
- **Module 6: creating and updating tickets via the ServiceNow Table API.** The Table API exposes /api/now/table/{tableName} and /api/now/table/{tableName}/{sys_id} with GET, POST, PATCH, PUT and DELETE; collection GETs return up to 10,000 records by default and sysparm_query accepts encoded queries for filtering and sorting. **[115]**

Agentic Attack Surface Management

- **Module: cloud inventory APIs (AWS Config).** AWS Config records the configuration and inter-resource relationships of AWS resources over time, and an aggregator can pull configuration and compliance data from many accounts and Regions into one place for a centralized inventory view. **[116]**
- **Module: cloud inventory APIs (Azure Resource Graph).** Azure Resource Graph lets you query resource properties at scale across subscriptions using a KQL-based language, returns only resources you have at least read RBAC access to, and keeps 14 days of resource configuration change history; its data is eventually consistent rather than strongly consistent. **[117]**
- **Module: Certificate Transparency logs as an asset-discovery source.** CT logs are publicly auditable, append-only, untrusted logs of issued certificates, designed so anyone can monitor CA activity and detect suspicious or unexpected certificates. **[118]**
- **Module: building a CT log monitor for new subdomains/certificates.** RFC 9162 defines the monitor role and requires that a monitor inspect at least every new entry in every log it watches, which is the same pattern an attack surface tool uses when watching CT logs for newly issued certificates on an organization's domains. **[119]**

- **Module: hands-on subdomain discovery via crt.sh.** crt.sh is a search front end over Certificate Transparency data that lets you look up certificates by identity (domain or organization name), by SHA-1/SHA-256 fingerprint, or by crt.sh ID, with an advanced search mode available. **[120]**
- **Module: DNS fundamentals and passive DNS.** DNS data is organized into zones, each authoritative for a connected region of the name space; each node holds a set of resource records whose order carries no meaning, and name servers may also cache non-authoritative data for other parts of the tree. **[121]**
- **Module: authorized scope and rules of engagement.** NIST defines Rules of Engagement as detailed guidelines and constraints for executing security testing, established before testing starts so the team has authority to perform defined activities without needing further permission; Appendix B supplies an ROE template. **[122]**
- **Module: CMDB concepts and entity resolution against the inventory.** A CMDB stores assets, services and the relationships between them as configuration items (CIs), which teams rely on to monitor infrastructure and keep it stable; external discovery results must be reconciled against these CIs. **[123]**
- **Module: reconnaissance and application enumeration on discovered assets.** WSTG section 4.1 lists ten information gathering tests, including search engine reconnaissance for information leakage, fingerprinting the web server, enumerating applications on a web server, identifying application entry points and mapping application architecture. **[124]**

Agentic Configuration Reviewer

- **Module 2: parsing Terraform plan JSON for review.** The plan JSON from `terraform show -json` carries a format_version ("1.0" since Terraform 1.1.0) and a resource_changes array whose change.actions values are no-op, create, read, update, delete, or the two replacement orderings ["delete","create"] / ["create","delete"], so a reviewer can detect any destructive change by checking for "delete". **[125]**
- **Module 3: Kubernetes YAML hardening checks against Pod Security Standards.** The Restricted profile builds on Baseline and requires allowPrivilegeEscalation: false, runAsNonRoot: true, a non-zero runAsUser, seccompProfile.type of RuntimeDefault or Localhost, capabilities.drop containing ALL with only NET_BIND_SERVICE allowed to be added, and restricts volumes to types such as configMap, secret, emptyDir, projected, csi, ephemeral and persistentVolumeClaim. **[126]**
- **Module 3: reading securityContext fields in Pod manifests.** A pod-level securityContext (runAsUser, runAsGroup, fsGroup, supplementalGroups) applies to every container in the Pod, while container-level fields such as allowPrivilegeEscalation apply only to that container; allowPrivilegeEscalation defaults to true when unset and cannot be false for privileged or CAP_SYS_ADMIN containers. **[127]**
- **Module 4: policy-as-code evaluation of Terraform plans with OPA/Rego.** OPA consumes the output of `terraform show -json` as `input`, and the tutorial policy walks resource_changes[].type and change.actions to compute a weighted blast-radius score, authorizing the plan only if the score is below 30 and no aws_iam resources change. **[128]**
- **Module 4: running Rego policies against Kubernetes/Terraform config files with Conftest.** Conftest evaluates Rego rules named deny, violation, and warn (optionally suffixed like deny_myrule) in the main namespace by default, against many structured formats including YAML, JSON, HCL/HCL2, Dockerfile, TOML, INI and XML. **[129]**

- **Module 4: scanner-based misconfiguration checks with Trivy.** `trivy config <dir>` auto-detects Terraform, Kubernetes, Dockerfile, CloudFormation, Azure ARM and Helm files and reports pass/fail counts per file by severity; with `trivy image` or `trivy fs` the misconfiguration scanner must be enabled explicitly via `--scanners misconfig`, and the checks bundle is downloaded from the internet unless an offline setup is used. **[130]**
- **Module 4: static IaC analysis with Checkov.** Checkov statically analyzes Terraform, CloudFormation/SAM, ARM, Serverless, Helm, Kubernetes and Docker files with more than 750 predefined policies, supports custom policies, and maps results to benchmarks such as CIS and the AWS Foundations Benchmark. **[131]**
- **Module 5: posting inline review findings via the GitHub pull request review comments API.** `POST /repos/{owner}/{repo}/pulls/{pull_number}/comments` requires body, commit_id, path and line (line is optional only when subject_type is file); `position` is being phased out in favor of line, multi-line comments use start_line/start_side, and the endpoint can trigger secondary rate limits when called rapidly. **[132]**
- **Module 5: reporting read-only check results and annotations via the Checks API.** Creating a check run (`POST /repos/{owner}/{repo}/check-runs`) requires a GitHub App, needs name and head_sha, supports conclusions such as success, failure, neutral and action_required, and limits annotations to 50 per request, with each annotation needing path, start_line, end_line, annotation_level and message. **[133]**

Agentic Architecture Reviewer

- **Module 1: STRIDE threat categories.** Microsoft's Threat Modeling Tool, part of the SDL, uses the STRIDE model of six threat categories (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) to turn vague security worries into pointed questions such as how an attacker could alter authentication data. **[134]**
- **Module 2: four-phase threat modeling workflow.** OWASP structures threat modeling around the Manifesto's four questions as phases: model the system (typically with data flow diagrams showing trust boundaries), identify and rank threats (STRIDE as the example technique), choose a response per threat (mitigate, eliminate, transfer, or accept), then validate that the model matches the real system and mitigations are testable. **[135]**
- **Module 2: values and principles for agentic reviews.** The Manifesto defines threat modeling as analyzing representations of a system to highlight security and privacy concerns, and values finding and fixing design issues over checkbox compliance, people and collaboration over process and tools, and continuous refinement over a single delivery. **[136]**
- **Module 3: C4 architecture abstractions.** The C4 model describes software at four hierarchical levels of abstraction, each with a matching diagram type: system context, containers, components, and code. **[137]**
- **Module 3: rendering C4 diagrams as Mermaid text.** Mermaid's C4 support is explicitly experimental and covers five diagram kinds (C4Context, C4Container, C4Component, C4Dynamic, C4Deployment) using syntax compatible with C4-PlantUML. **[138]**
- **Module 3: embedding Mermaid diagrams in review output.** GitHub renders Mermaid diagrams written inside a fenced code block tagged with the mermaid language identifier, and a code block containing just 'info' reveals which Mermaid version GitHub is running. **[139]**
- **Module 4: documenting decisions with ADRs.** An Architectural Decision Record captures a single architectural decision together with its rationale, so readers can later understand the

trade-offs and consequences behind the choice. **[140]**

- **Module 5: Terraform plan output as review evidence.** terraform plan previews changes without applying them; the -out flag saves a plan file (which contains configuration and planned values in cleartext and must be treated as sensitive) and the -json flag emits machine-readable JSON UI output, which also implies -input=false. **[141]**
- **Module 5: Kubernetes evidence for findings.** Kubernetes defines three cumulative Pod Security Standards: Privileged (unrestricted, for trusted infrastructure workloads), Baseline (blocks known privilege escalations such as host namespaces, privileged containers, and hostPath volumes) and Restricted (follows current pod hardening best practices). **[142]**

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